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Magnetic polarity stratigraphy of Inchasi: a Pliocene mammal-bearing locality from the Bolivian Andes deposited just before the Great American Interchange

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ABSTRACT

The newly discovered section at Inchasi, located about 50 km southeast of Potosi, Bolivia, in the eastern Cordillera, consists of about 120 m of undeformed terrestrial sediments containing fossil mammals. Paleomagnetic analysis of 54 sites indicates a polarity pattern with an estimated duration of about 0.64 Ma. The rich Inchasi local fauna indicates a Montehermosan and/or Chapadmalalan land mammal age (Pliocene). Given these constraints, Inchasi correlates from the interval between the late Gilbert (within the Cochiti subchron) to the early Gauss (within the Mammoth subchron) chrons; that is, between about 4.0 and 3.3 Ma. The distinct lack of North American mammals in the Inchasi local fauna provides some of the first well-calibrated evidence that the Great American Interchange occurred after about 3.0 Ma, as has been previously stated based on other calibrations of the earliest immigrant (Uquian) faunas.

1. Introduction

During most of the Cenozoic, South America drifted as an isolated continent carrying a unique biota. This isolation and faunal endemism changed after the formation of the Isthmus of Panama and the corresponding influx of immigrant taxa from North America during the late Cenozoic, resulting in what has been termed the "Great American Interchange" [1,2]. This interchange occurred in several phases, beginning in the late Miocene, about 8–9 myr ago, but the major phase of mammalian dispersal occurred during the Pliocene and Pleistocene, during the Uquian, Ensenadan, and Lujanian land mammal ages. These biochrons are relatively well calibrated at certain classic localities [3–8], however, in contrast, precise age determinations of pre-interchange faunas and mammal ages in South America are relatively infrequent.

The basin deposits of the central Andes of Bolivia preserve a rich sequence of mammalian faunal evolution for much of the Cenozoic. Having been deposited in a tectonically active setting, these basins often contain long, continuous stratigraphic sections of the kind amenable to magnetostratigraphic analyses. Over the past few years, we have studied and sampled the late Cenozoic sediments surrounding the village of Inchasi in eastern Bolivia. Our collections have revealed a relatively rare pre-interchange mammalian fauna. These discoveries, in conjunction with new magnetostratigraphic data presented for the Inchasi beds, allow a better understanding of the endemic mammals and precise calibration of the Pliocene of the central Andes. In a broader context, the data from Inchasi further refine and calibrate the phase of South American geological history just prior to the major pulse of faunal interchange during the late Cenozoic Great American Interchange.

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2. Geological setting and relevant studies

The Communidad Inchasi, at an elevation of about 3220 m, is located 8.5 km to the northeast of Puna and 50 km to the southeast of the city of Potosi in the Province of Linares, Department of Potosi at latitude 19°43′15″S, longitude 65°28′6″W (Fig. 1). Pliocene sediments, referred to here as the Inchasi beds, were previously mapped as Quaternary [9]. They are represented by a 120 m thick section of horizontally bedded sediments (Fig. 2) that outcrop in a structural basin, the margins of which are formed by Ordovician sediments and the prominent middle Miocene Kari Kari caldera (previously thought to be a batholith, e.g., [10]). Within the Inchasi beds the dominant lithologies are high-energy fluvial gravels interbedded with finer-grained clays and silts. There are several tuffaceous units, although results of radioisotopic analysis in the laboratory (see below) indicates that they are of secondary detrital origin. As also discussed below, numerous fossiliferous horizons are present at Inchasi.

During our study we measured five stratigraphic sections, which contained fossils and were potentially datable, using magnetostratigraphy and radioisotopic age determinations. The field relations and correlation of four of these (sections 1, 2, 3A and 3B) are easily demonstrable because they are located in close proximity to one

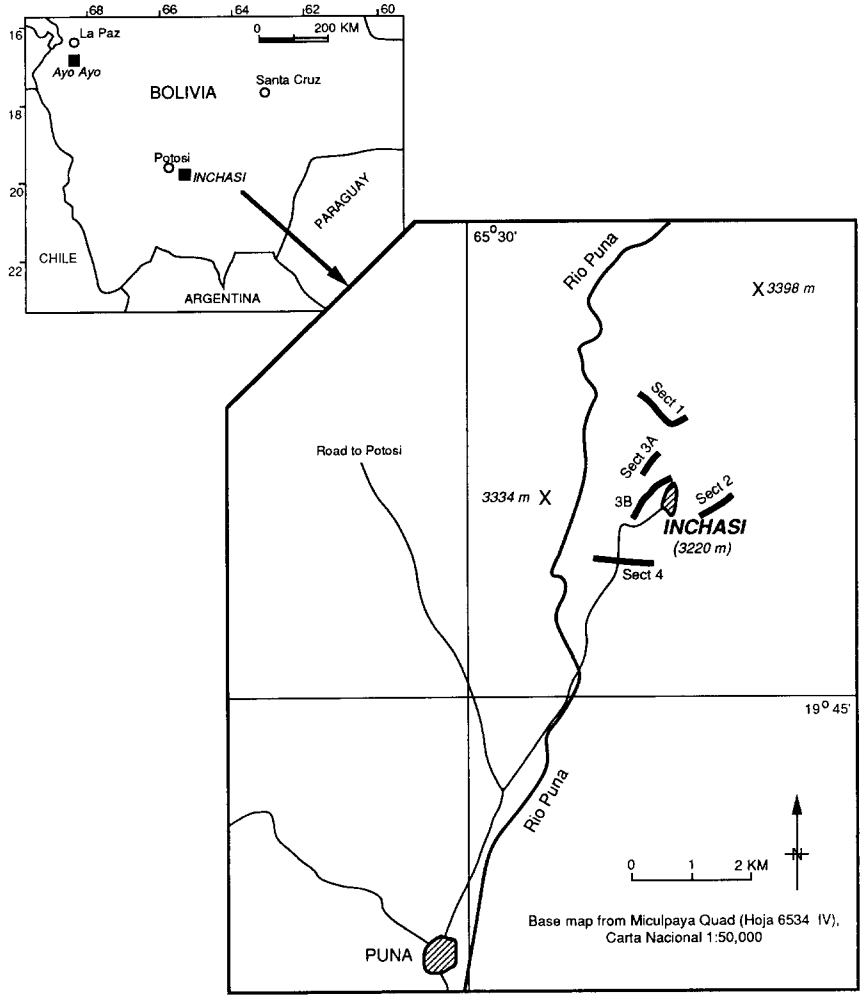


Fig. 1. Map of the Inchasi region showing location of the five stratigraphic sections measured during this study (see also Fig. 2). Inset: Location of Inchasi in Bolivia and relative to another site (Ayo Ayo) discussed in text.

another (1 km or less) and, in many cases, important marker beds can be traced from one section to the next. Another important section, no. 4, is located to the southwest and correlation to the others is difficult, other than from its elevation and general position in the section.

3. Paleomagnetic sampling, laboratory analysis and results

Following standard procedure, a minimum of three separately oriented samples were collected in the field from each of the 54 sites of suitable fine-grained lithologies throughout the five measured sections (1, 2, 3A, 3B and 4) within the Inchasi beds. Although two of the longer sections (1 and 2; see Fig. 2) contain massive conglomerates, there were sufficient interbedded fine-grained units that were amenable to paleomagnetic sampling.

In the laboratory these samples were prepared to standard-sized 2.5 cm cubes and then measured in a SCT cryogenic magnetometer and treated in a Schonstedt alternating field (AF) and thermal demagnetizers in the University of Florida Paleomagnetics Laboratory. During analysis, these specimens were stored in the magnetically shielded laboratory. Samples for IRM saturation studies were magnetized in high fields and then measured on a Schonstedt spinner magnetometer.

IRM saturation experiments: In order to better understand the magnetic mineralogy, eight samples representing the varied lithologies of the Inchasi beds were analyzed for saturation magnetization. As represented in Fig. 3A, in the majority of cases, saturation occurs in relatively low fields of 1 T (Tesla) or less, suggesting that magnetite is the dominant carrier of the NRM. Some other samples, however, show increasing magneti-

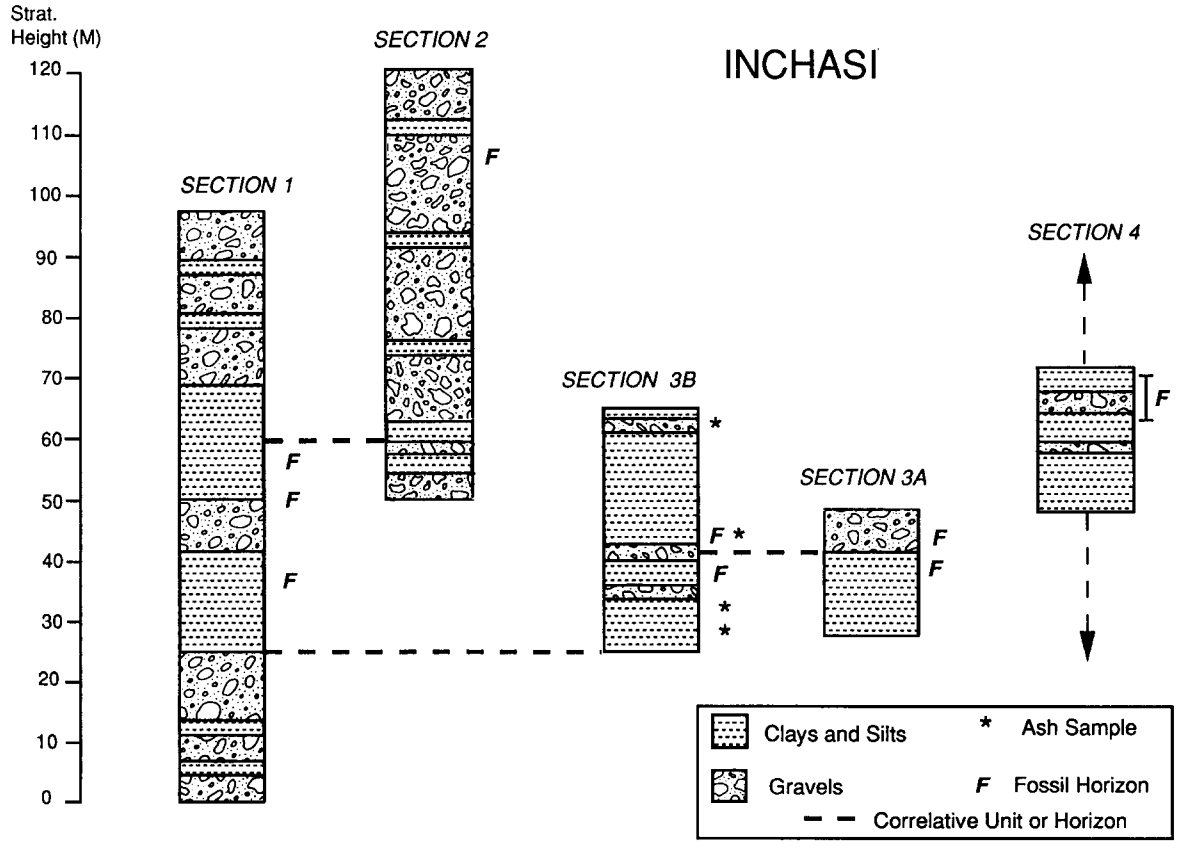


Fig. 2. Stratigraphic sections at Inchasi measured during this study, with the dominant lithologies represented, correlations between sections and position of fossil and ash sample horizons. (For position of paleomagnetic sites, see Fig. 7).

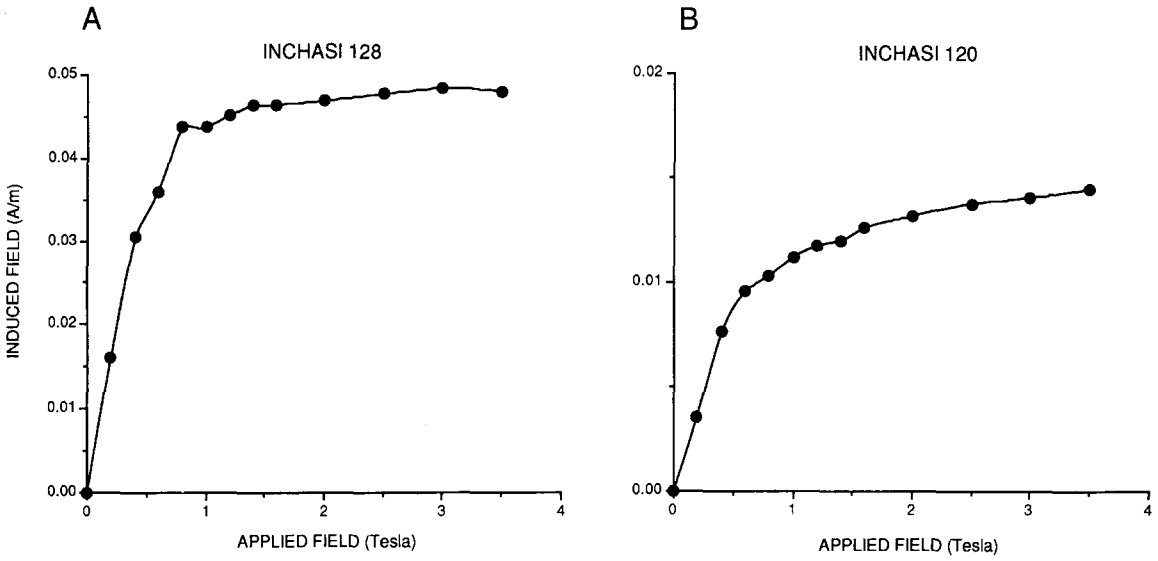


Fig. 3. Isothermal remanence curves for two samples from Inchasi. (A) A sample in which saturation occurs in low fields. (B) A sample in which the induced field increases with increasing applied field. (A) is characteristic of low-coercivity magnetic minerals, probably magnetite, whereas (B) is characteristic of high-coercivity minerals, probably hematite and/or goethite.

zation in greater applied fields exceeding 3 T (Fig. 3B). These latter results suggest that hematite or geothite are the dominant magnetic mineral(s), as are also indicated from thermal demagnetization results presented below.

Paleomagnetic measurements and analysis: One sample from each site was subjected to a uniform regime of AF demagnetization (usually 15 steps between 5 and 99 mT). The remaining two samples from each site were subjected to uniform regimes of thermal demagnetization (usually 12–20 steps between 100° and 650°C). Furthermore, the AF demagnetized samples were also subsequently thermally demagnetized to remove any high coercivity or high blocking temperature components not affected by alternating fields.

In many cases the AF demagnetization regime revealed that the remanent magnetization was relatively straightforward, with stable components decaying towards the origin after removing a low coercivity, presumably viscous, magnetization (Fig. 4A and B). The possible presence (as determined from the IRM experiments) of high coercivity minerals; that is, hematite and/or goethite, dictated that the AF treatment was not sufficient and therefore all samples were also thermally demagnetized. With this technique it was found that, as with the AF, the magnetization is relatively straightforward in most cases. After removal of a low blocking temperature component (usually by 300°C or less), the demagnetization at higher temperatures (up to 650°C) revealed a stable component, carried in what are interpreted as the magnetite and hematite phases of the remanent magnetization (Fig. 4C). Other samples indicate single-component behavior, with little change in direction upon demagnetization, even at low temperatures (Fig. 4D and E). The presence of both normal and reversed sites (see below), and reversals upon further demagnetization, as well as stable component decay, indicates that the remanence of the Inchasi beds is relatively simple and, therefore, can be used to interpret site polarities, as will be discussed below.

Results: The mean NRM intensity of all measured samples is 2.62×10^{-4} A/m ($N = 162$). After partial step-wise demagnetization, intensities averaged, for example, 0.39×10^{-4} A/m at 50 mT ($N = 44$) and 0.64×10^{-4} A/m at 500°C ($N = 162$). Principal components analysis [11] was carried out on all samples to obtain the least-squares, best fit directions of the decay of the characteristic magnetic component. Most final paleomagnetic directions were obtained in this manner; however, there were a few samples with poorly constrained principal components (i.e.,

MADs > 20). For these, the final directions were chosen from a single-step, usually thermal, demagnetization between 400° and 575°C, which were interpreted as representing the characteristic magnetization. For each site the individual sample data were then combined using standard statistics [12] to obtain mean directions and associated parameters.

As shown in Fig. 5, after demagnetization and statistical analysis, relative to the untreated (NRM) data, the site means grouped into two populations, with 50 of the 54 originally sampled sites being Class I data [13]. This relatively high number of Class I sites (93%) resulted because the characteristic magnetization was either straightforward or easily isolated through demagnetization. The mean of all normal sites is dec. 358.2°, −22.0° ($N = 11$, $\alpha_{95} = 8.6°$, $k = 29.1$), of all reversed sites is dec. 175.7°, inc. 33.8° ($N = 39$, $\alpha_{95} = 6.7°$, $k = 12.6$), and of all sites (reversed inverted) is dec. 356.3°, inc. −31.13° ($N = 50$, $\alpha_{95} = 5.6°$, $k = 13.8$). The difference in the normal and reversed site mean directions probably reflect incomplete removal in the laboratory of a secondary overprint.

The sediments at Inchasi are horizontally bedded, so no fold test could be performed on the paleomagnetic data. However, the reversals test

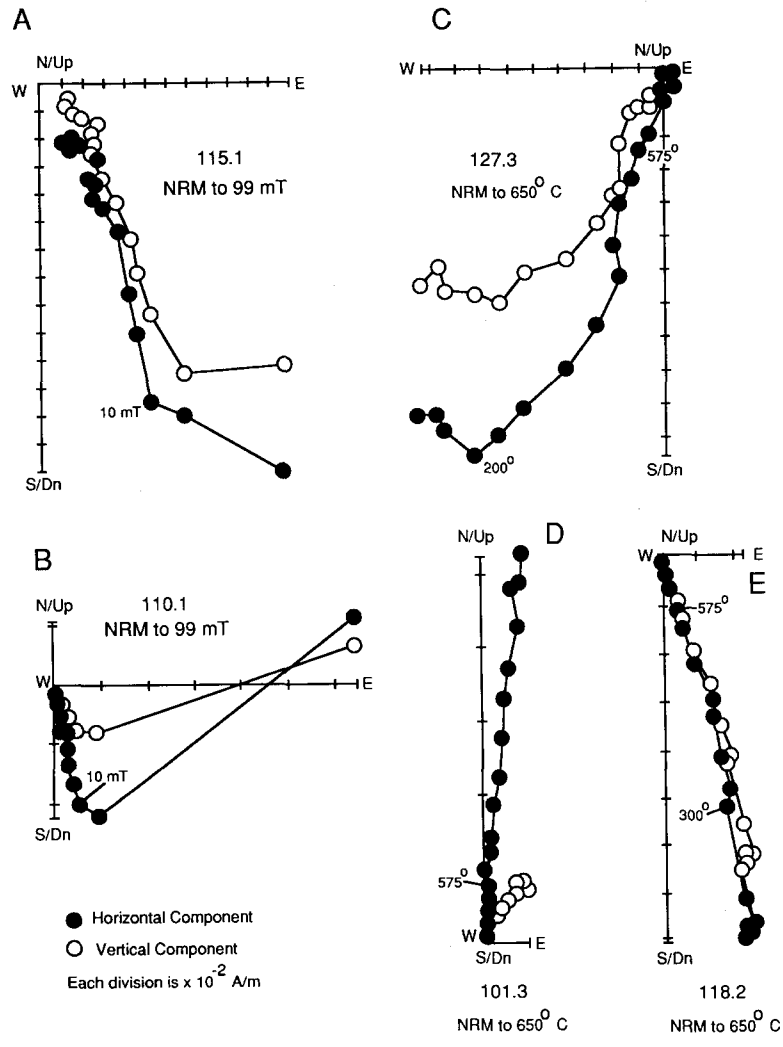


Fig. 4. Orthogonal vector demagnetization ("Zijderveld") diagrams showing the behavior of magnetic vector components (horizontal and vertical) upon alternating field (A, B) and thermal (C–E) demagnetization.

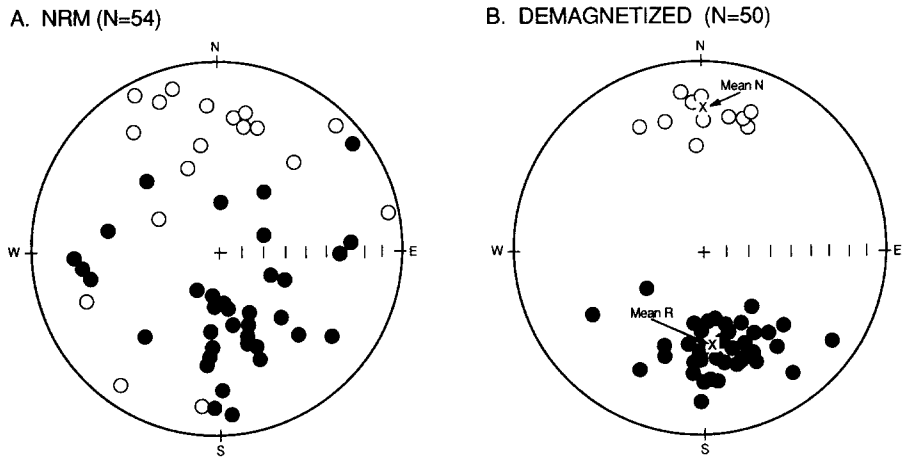


Fig. 5. Site mean directions plotted on equal-angle stereographic projections. (A) Before demagnetization for all 54 sites collected. (B) After demagnetization for the 50 Class I sites.

(Fig. 6), with two overlapping α_{95} cones of confidence, show that there are two antipodal populations interpreted as being of normal and reversed polarity. In summary, the site mean directional data, roughly antipodal N and R populations, and the overlap of the α_{95} cones of confidence, indicate that the magnetization of Inchasi beds are stable and record the polarity of the ancient dipole field at, or near, the time of deposition.

4. Comments on radioisotopic age determinations

As mentioned above, during the present study tuffaceous samples were collected in the field by Carl C. Swisher for possible radioisotopic age determinations. These samples were taken from four separate horizons within section 3B (Fig. 2). Minerals were separated from these ashes and analyzed at the Berkeley Geochronology Laboratory using the $^{40}Ar/^{39}Ar$ method. These results, which are not reported here, all yielded middle Miocene ages similar to those previously reported for the surrounding Kari Kari caldera [10]. Thus, the radioisotopic data from our analyses represent older detrital contamination and, as such, do not determine the age of deposition of the Inchasi beds.

5. Paleontology and biostratigraphy

Although the altiplano and adjacent cordillera of the central Andes have a relatively rich se-

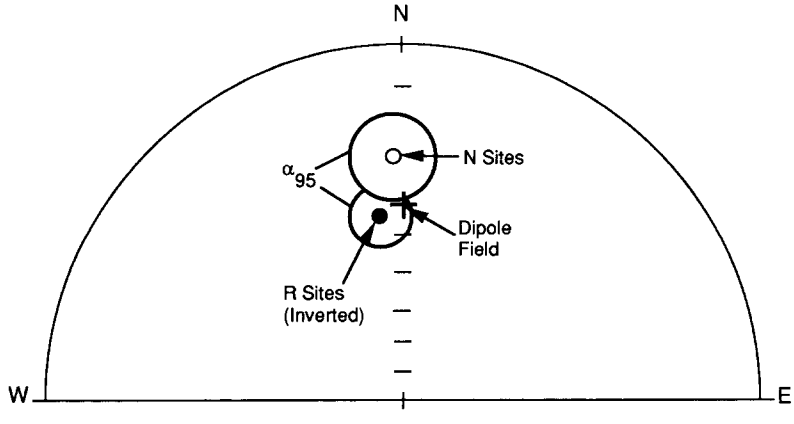


Fig. 6. Reversals test showing overlap of normal and reversed (inverted to northern hemisphere) sites, corresponding α_{95} cones of confidence and dipole field.

quence of described mammalian faunas [14], the Inchasi locality was unknown until this study. Fossil mammals were first collected by two of us (MacFadden and Argollo) during a reconnaissance to Inchasi in 1989. Subsequent field work has revealed that this basin is richly fossiliferous, with important specimens coming from all of our measured sections (Fig. 2). Generally, the condition of the fossil specimens range from isolated teeth and skeletal elements to partially articulated individuals. A detailed description of the mammalian fauna from Inchasi is currently being compiled by one of us (Anaya). The brief discussion that follows here is intended to provide a biostratigraphic age assessment for correlation to the MPTS (Magnetic Polarity Time Scale).

Although the fauna recovered so far is not particularly diverse (Table 1), and further collecting would undoubtedly add newly discovered taxa, there are some general patterns that are of importance in a biostratigraphic context. The most striking aspect of the fauna is that it lacks any characteristic North American component and, as such, is distinctly pre-interchange in age. Based on this knowledge, the fauna is older than Uquian land mammal age and, therefore, older than about 2.5 Ma [5,7].

As can be seen from the identifications in Table 1, lack of both recent taxonomic revisions and detailed comparisons with classic localities in the late Cenozoic of Argentina hamper species-level identifications and, hence, more precise stage of evolution age assessment for the Inchasi local fauna. Nevertheless, several of the genera are important in constraining the age of the Inchasi local fauna. According to the biochrono-

TABLE 1

Fossil mammals encountered at Inchasi

Order EDENTATA Cuvier, 1798
Suborder XENARTHRA Cope, 1889
Infraorder PILOSA Flower, 1883
Family MEGATHERIIDAE Owen, 1843
Proscelidodon patrius (Ameghino, 1888), **Mt–Ch**
Family MYLODONTIDAE Gill, 1872
Megatheriinae, indeterminant
Glossotheridium cf. *chapadmalensis* (Kragliev., 1925), **Ch**
Mylodontinae, indeterminant
Infraorder CINGULATA Illinger, 1811
Family PAMPATHERIIDAE Paulo Couto, 1954
Plaina sp., **Mt–Ch**
Family GLYPTODONTIDAE Burmeister, 1879
Plohophorus sp., **Mt–Ch**
Paraglyptodon sp., **Ch**
Order RODENTIA Bowdich, 1821
Suborder CAVIOMORPHA Wood and Patterson 1955
Family CAVIIDAE Waterhouse, 1839
Caviodon sp., **?Mt–Ch**
Family HYDROCHOERIDAE Gill, 1872
Chapalmatherium cf. *saavedrai*., **Ch**
Order LITOPTERNA Ameghino, 1889
Family MACRAUCHENIIDAE Gill, 1872
Promacrauchenia sp., **Mt–Ch**
Order NOTOUNGULATA Roth, 1903
Family TOXODONTIDAE Gervais, 1847
Posnanskytherium desaguaderoi Liendo, 1943, **Ch**
Posnanskytherium sp., **?Mt–Ch**
Family MESOTHERIIDAE Alston, 1876
Mesotheriidae, new genus

The abbreviations Mt and Ch refer to the range of the taxon according to Pascual [15] and Marshall et al. [16].
Mt = Montehermosan (early to middle Pliocene); **Ch** = Chapadmalalan (late Pliocene).

logic ranges in Pascual [15] and Marshall et al. [16], none of the genera at Inchasi are restricted to the Montehermosan (*sensu stricto*; that is, excluding the Chapadmalalan), whereas *Glossotheridium*, *Paraglyptodon* and *Chapalmatherium* are characteristic of the Chapadmalalan (although the latter taxon may originate in the Montehermosan *s. l.*, according to Marshall et al. [16]). Longer-ranging taxa, known to occur in both the Montehermosan and Chapadmalalan, include *Proscelidodon*, *Plaina*, *Plohophorus*, *Promacrauchenia*, and possibly *Caviodon* and *Posnanskytherium*. Interestingly, according to Marshall et al. [16] none of these taxa extend to older (Huayquerian = late Miocene) or younger (Uquian = late Pliocene/early Pleistocene) ages. Only the new genus of mesotheriid toxodont may represent a local range extension in Bolivia (it persists in Argentina [16]), but the further significance of that discovery must await formal description of it (Anaya, in preparation). As such, the Inchasi local fauna seems to indicate an undifferentiated Montehermosan/Chapadmalalan age. It is not surprizing that this local fauna contains a mixture of taxa characteristic of these two biochrons. As discussed in several studies [e.g., 16], for a long time these two land mammal ages have been difficult to distinguish. If they are, indeed, distinct, then differentiation will require a more precise stage of evolution biochronology (probably including species-level discrimination) than that at present available at Inchasi. In summary, with the biochronological data now available, the Inchasi local fauna is characteristic of the early to middle Pliocene and correlates to the classic faunas of Montehermosan and/or Chapadmalalan age in Argentina (see also discussion below).

Pliocene faunas from elsewhere in the Bolivian Andes and altiplano are less well known, and

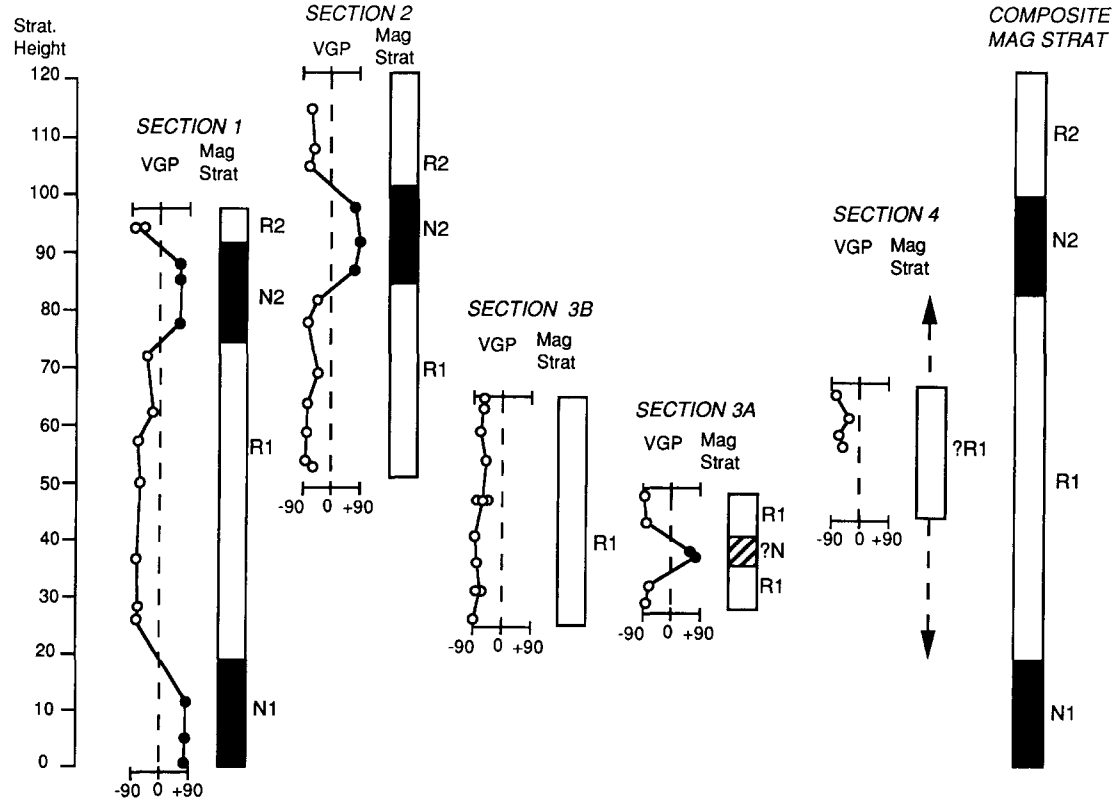


Fig. 7. Magnetic polarity stratigraphy of the five measured sections and composite section at Inchasi. The relative positions of sections 1, 2, 3A and 3B are determined from lithostratigraphic correlations (see Fig. 2). As discussed in the text, the position of section 4 is based on an estimate of its general position in the middle of the composite stratigraphic column.

relative to Argentina, there is regional endemism of certain taxa [14,16]. Inchasi appears to be correlate closely with Ayo Ayo [17,18], a Pliocene sequence of predominantly lacustrine sediments located south of La Paz (Fig. 1). Although there are some independent geochronological data points to calibrate the classic Pliocene faunas from Argentina [16], until recently, none were available for those of similar age in Bolivia. Accordingly, the study of the magnetostratigraphy of the Umala Formation at Ayo Ayo and correlative units is relevant here. Thouveny and Servant [19] found that at Ayo Ayo the polarity was dominantly reversed (2 normal sites, 16 reversed sites) and it was correlated to the Gilbert chron (revised ages between 3.6 and 5.7 Ma using Cande and Kent [20]). This correlation is further corroborated by a recently reported $^{40}Ar/^{39}Ar$ age of 2.9 ± 0.006 Ma for the Ayo Ayo tuff at the top of the section [21]. These data are consistent with the present study and will be used in the correlation below of the magnetostratigraphy at Inchasi to the MPTS.

6. Magnetostratigraphy

Figure 7 depicts the position of paleomagnetic sites and interpreted magnetic polarity zones throughout the five measured sections. Section 1, which is the longest, contains four polarity zones N1–R2. In this section, zone R2 was originally based on a single site, however, additional resampling of a laterally correlative horizon during subsequent field work, and the presence of this zone in section 2, corroborates that it is indeed valid. Based on lithostratigraphic correlations, the three polarity zones represented in section 2 correlate to R1–R2. In sections 3A and 3B the dominant polarity is reversed and, based on field observations of their stratigraphic position (Fig. 2), these are correlated to R1. In section 3A there is a small zone of normal polarity represented by two sites. However, given the fact that this zone does not extend laterally to any of the other local sections, it is considered that this must either: (1) be a local occurrence within a small zone of higher sedimentation (relative to the other sections) or non-erosion; or (2) have resulted from spurious magnetization. With regard to section 4, so far as is determined (the bottom part of this section has poor magnetic results), the four sites of reversed polarity in the top half of this section probably correlate to R1. However, it should be noted that, because it is located to the southwest of the other sections (Fig. 1), and individual lithostratigraphic units cannot be traced, this is only a tentative correlation. In summary, four polarity zones N1–R2 are represented in the composite magnetostratigraphic section at Inchasi (Fig. 7).

7. Correlation to the magnetic polarity time scale

As mentioned above, unfortunately the radioisotopic age determinations did not yield results that bear on the age of the Inchasi sediments. Nevertheless, there are several other lines of evidence that, when taken together, constrain the correlation of the local section at Inchasi to the Magnetic Polarity Time Scale (MPTS).

Biostratigraphy: As described above, the biochronology of the Inchasi local fauna is clearly of Montehermosan/Chapadmalalan land mammals ages, or if just the Montehermosan is considered valid, Montehermosan *s. l.* This age is equivalent to middle Pliocene, or about 5–2.5 Ma [16]. Furthermore, the distinctive lack of characteristic North American immigrants also indicates an age prior to about 2.5 Ma [5]. Using the revised MPTS [20], correlation of the magnetic section at Ayo Ayo indicates a Gilbert age between about 5.7 and 3.6 Ma. As mentioned above, the upper limits of this correlation are further constrained by the $^{40}Ar/^{39}Ar$ age of 2.9 ± 0.006 Ma [21].

Temporal duration: If the local section being studied is adequately sampled for magnetostratigraphy, then the number of reversals encountered can provide an estimate of the amount of time represented [22]. The sampling density at Inchasi is 16.7 per reversal, which is considerably greater than that recommended by Johnson and McGee [22] (> 8) to sample the section adequately. Using the equation from [22], the duration, Δt, is calculated by:

$$\Delta t(\text{myr}) = S\tau N$$

where: $S = -\ln(1 - 2p)/2$; τ = the individual duration (time length) of polarity zones; N = number of local paleomagnetic sites sampled; p

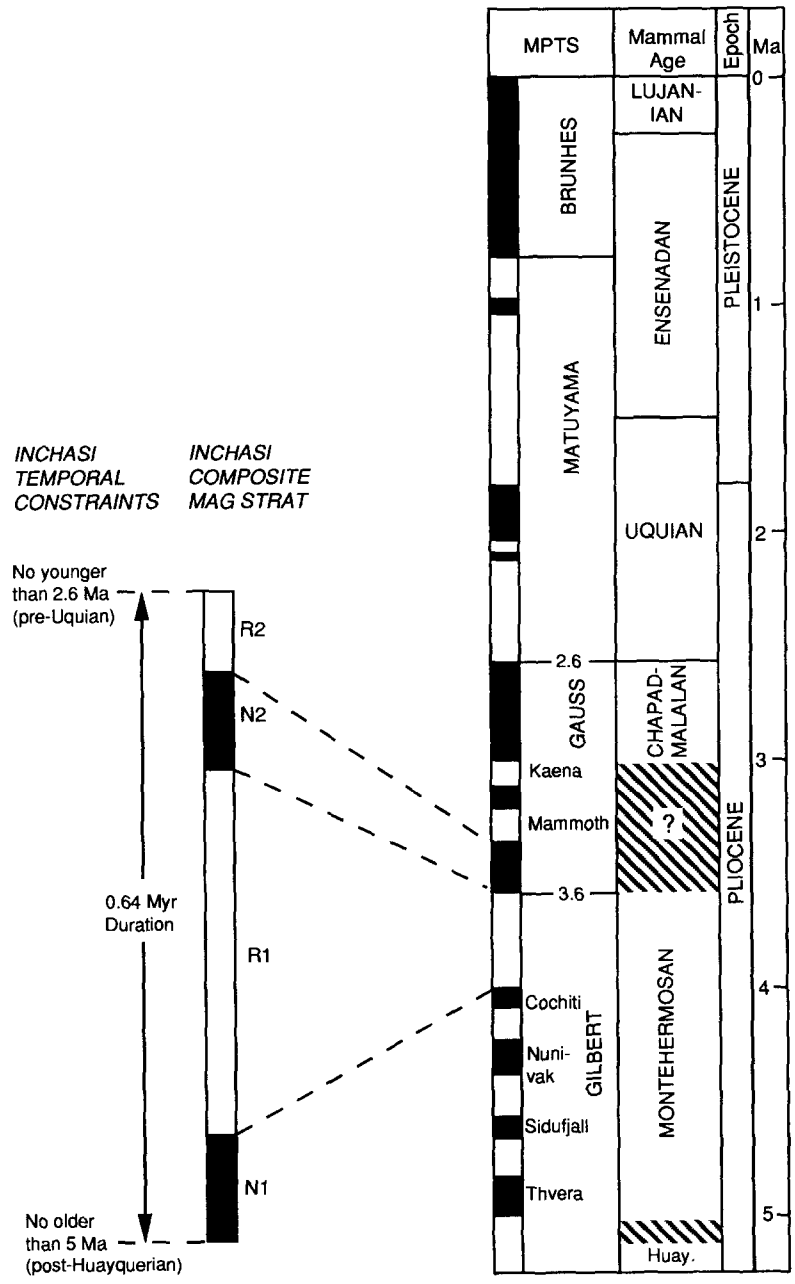


Fig. 8. Correlation of the Inchasi composite magnetostratigraphy to the time scale (MPTS) given: (1) its polarity pattern; (2) estimated duration (0.64 myr); and (3) constraints from the presence of Montehermosan/Chapadmalalan mammals. Thus, using the time scale of Cande and Kent [20], in the preferred hypothesis (see discussion in text), this section correlates to the late Gilbert and early Gauss magnetic chrons and has an age range of between about 4.0 and 3.3 Ma. Note the following: (1) the South American land mammal age boundaries follow Marshall et al. [1,7,16], but are adjusted, so far as is possible, to the revised MPTS [20]; (2) although Cande and Kent [20] use alpha-numeric chron designations, the widely recognized chron/subchron names (e.g., Gauss/Kaena) are used above; (3) the recognition of Chapadmalalan as a distinct land mammal age is controversial and it may represent the later part of Montehermosan time.

$= R/(N-1)$, where R is number of reversals encountered.

The values for N, R, P and S are all derived from the paleomagnetic results at Inchasi. Using the time scale of Cande and Kent [20] and simply counting the number of major (i.e., excluding the "tiny wiggles," or cryptochrons of less than 30 kyr duration) reversals (48) over the last 10 Ma, the value of τ is taken to be 0.21 Ma. Substituting these values yields an estimate of 0.64 myr for the amount of time in which the sedimentary section at Inchasi was deposited. Although this value, by itself, does not uniquely constrain the correlation to the MPTS, this estimate of sedimentary duration will contribute to the overall correlation along with the other lines of evidence presented here.

Correlation: The composite section at Inchasi (Fig. 7) consists of four magnetozones (N1–R2) of which R1 is the longest. In conjunction with the constraints that the section should fall between about 5.7 and 2.5 Ma and span about 0.64 myr, the correlation made here (Fig. 8) is to the upper Gilbert through lower Gauss, which spans an age from 4.0 to 3.3 Ma. This is our preferred hypothesis, but we also realize that it is not the only correlation that can be made. Two other possibilities are as follows:

(1) At the older end of the spectrum, the Inchasi beds could be correlated to the more complex part of the early and middle Gilbert chron, which includes the Thvera to Cochiti subchrons. This correlation would require rejection of the estimated duration (0.64 myr) calculated above in favor of a shorter time span of about 0.3–0.4 myr. In addition, this older correlation seems less likely because of the predominance of Chapadmalalan mammals in the Inchasi locla fauna.

(2) At the younger end of the spectrum, and also assuming a shorter duration, it cannot be ruled out that the Inchasi beds correlate to the middle of the Gauss chron. Due to the predominant length of R1 at Inchasi, however, this latter hypothesis requires an assumption of non-constant sedimentation, because the Kaena and Mammoth subchrons, both of which are reversed, are shorter than the normal subchrons within the Gauss.

Either of these two (and other possible) correlations require that one or more of the several lines of evidence presented above for the preferred hypothesis must be considered spurious. Given the requirement of these ad hoc assumptions for the other hypotheses, we assert that our correlation represents the best hypothesis given the available data. Thus, our data indicate that the Inchasi beds, and the fossil mammals contained within it, are of middle Pliocene age, correlate from the Cochiti subchron within the late Gilbert chron to the Mammoth subchron within the earliest Gauss chron and are about 3.3 to 4.0 myr old.

The correlation presented here for Inchasi is also consistent with magnetostratigraphic studies of the classic and better known sites in Argentina. Orgeira and Valencio [23] studied the Pliocene–Pleistocene sequence exposed at Barranca de Los Lobos, south of the city of Mar del Plata. The lower part of that sequence is represented by the Chapadmalal Formation, containing fossil mammals of typical Montehermosan/Chapadmalalan age. Their paleomagnetic results indicate that the Chapadmalal Formation is dominantly reversed, which they correlate to the upper Gilbert chron. Using the revised calibrations of the MPTS [20], this correlation spans the interval from about 5 to 3.6 Ma. Both the general magnetic pattern (dominantly reversed, although with several normal subchrons) and the correlation is thus comparable to that of Inchasi.

The next younger land mammal age (Uquian), is also well constrained by magnetostratigraphic correlations in Argentina and, as such, provides an upper age constraint for the Montehermosan/Chapadmalalan. As mentioned above, Marshall et al. [5] found that the Uquia Formation of Jujuy Province is also dominantly reversed. With revised calibrations [20] this correlates to the early Matuyama chron between about 2.6 and 1.5 Ma. Ogeira [8] studied another Uquian sequence, the Varohué and San Andreas formations, exposed at "El Muelle" from the Atlantic coastal area, Province of Buenos Aires. She concluded that these formations were of Gauss age, and thus older than about 2.6 Ma [20], which if correct, may be slightly older than that reported for the Uquia Formation above. Nevertheless, both of these studies constrain the Uquian land mammal age, include the earliest interchange faunas and are younger than the Montehermosan/Chapad-

malalan, late Gilbert/early Gauss age for the Inchasi beds in Bolivia.

8. Conclusion

With the addition of the magnetostratigraphy and chronology at Inchasi and correlations with other important localities in Bolivia and Argentina, the calibrations of the South American land mammal ages for the last 5 myr are now fairly well constrained. As shown in Fig. 8, Montehermosan/Chapadmalalan faunas should occur within sediments deposited during the Gilbert and Gauss chrons. Due to the many subchrons within this interval, without precise geochronological control, unique correlations of localsections to the MPTS are difficult. Following Marshall et al. [16,20], Uquian faunas fall within the early Matuyama chron, are about 2.6–1.5 Ma, and record the first distinctive North American component of immigrant mammals. With the preferred correlation presented above, spanning an age of between 4.0 and 3.3 Ma, the middle Pliocene Inchasi beds represent one of the few well calibrated localities in South America deposited just before the Great American Interchange.

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